

Graph Theory and Applications

Session 1 — From an investigation to our first theorems

Master 1 APE

How we shall work. The course is organised as a sequence of problems. New definitions will be introduced only when they become useful. Every result proved today will then belong to our common mathematical toolkit and may be used later without being proved again. Work in small groups. A drawing or a numerical observation is a useful starting point, but every general claim must ultimately be justified.

1 Who killed the Duke of Densmore?

Adapted from Claude Berge, Qui a tué le duc de Densmore?, Bibliothèque Oulipienne no. 67 (1994).

One day, Sherlock Holmes received a visit from his friend Watson, who had been asked to reopen a mysterious murder case dating back more than ten years.

The Duke of Densmore had been killed by the explosion of a bomb that destroyed Densmore Castle, where he had retired. His will, also destroyed by the explosion, was said to have greatly displeased his seven former wives. Shortly before his death, the Duke had invited all seven of them to spend a few days at his Scottish castle.

Holmes: I remember the case. The bomb had been specially made to fit inside a suit of armour in the Duke's bedroom. The murderer must therefore have visited the castle several times.

Watson: That is precisely why I questioned the Duke's former wives: Ann, Betty, Charlotte, Edith, Felicia, Georgia and Helen. Each of them swore that she had visited Densmore Castle only once in her life.

Holmes: Did you ask them when they stayed there?

Watson: Unfortunately, none of them could remember the exact dates. I did, however, ask whom they had met at the castle. Since I could compare their answers, they could not safely lie about a meeting:

- Ann met Betty, Charlotte, Felicia and Georgia;
- Betty met Ann, Charlotte, Edith, Felicia and Helen;
- Charlotte met Ann, Betty and Edith;
- Edith met Betty, Charlotte and Felicia;
- Felicia met Ann, Betty, Edith and Helen;
- Georgia met Ann and Helen;
- Helen met Betty, Felicia and Georgia.

Holmes drew seven points labelled A, B, C, E, F, G, H , joined some pairs of points, and announced that the information uniquely identified the murderer: exactly one woman had lied and visited the castle more than once.

Part I — Turning testimony into a mathematical object

- 1.1.** Holmes uses one labelled point for each woman. Decide when two points should be joined by a line. Draw the resulting diagram from the seven testimonies. At this stage, use only the meaning of the story: no mathematical vocabulary is required.

A finite simple undirected graph

A *finite simple undirected graph* is a pair $G = (V, E)$, where V is a finite set of *vertices* and

$$E \subseteq \{\{u, v\} : u, v \in V \text{ and } u \neq v\}.$$

The elements of E are the *edges* of G .

The condition $u \neq v$ forbids an edge from a vertex to itself; such an edge would be called a *loop*. Since E is a set, the same pair $\{u, v\}$ cannot occur twice; thus there are no *parallel edges*. The adjective *simple* refers precisely to the absence of loops and parallel edges. The graph is *undirected* because an edge is an unordered pair: $\{u, v\} = \{v, u\}$, unlike the ordered pair (u, v) .

The number $n = |V|$ is the *order* of the graph, and $m = |E|$ is its *size*.

In a drawing of G , every vertex is represented by a point and every edge $\{u, v\}$ by a line joining the points u and v . The positions of the points and the shapes of the lines are not part of the graph. A crossing of two lines is not a new vertex unless a labelled point is explicitly drawn there.

- 1.2.** Let $G = (V, E)$ denote the graph described by the testimonies. Write V and E explicitly. Then determine its order n and its size m .

Adjacency, incidence, neighbourhood and degree

Let $G = (V, E)$ be a finite simple undirected graph.

- Two distinct vertices u and v are *adjacent* when $\{u, v\} \in E$.
- If $e = \{u, v\} \in E$, then u and v are the *endpoints* of e . The edge e is *incident with* each of its endpoints.
- The *neighbourhood* of a vertex v is the set

$$N(v) = \{u \in V : \{u, v\} \in E\},$$

consisting of all vertices adjacent to v .

- The *degree* of v , denoted by $d(v)$, is the number of neighbours of v :

$$d(v) = |N(v)|.$$

A vertex of degree 0 is called an *isolated vertex*.

- 1.3.** For every vertex of the Densmore graph, write its neighbourhood and compute its degree. Translate the meaning of $d(v)$ back into the language of the investigation.
- 1.4.** Draw G again, but place the seven points in substantially different positions. Explain precisely why your two drawings represent the same graph.

A graph can also be encoded without drawing it. First choose an ordering of the vertices, written $v_1, \dots, v_n$.

Adjacency matrix

An $n \times n$ matrix is a square table with n rows and n columns. Its entry in row i and column j is denoted by a_{ij} .

The *adjacency matrix* of G , for the chosen ordering $v_1, \dots, v_n$, is the $n \times n$ matrix $A = (a_{ij})$ defined by

$$a_{ij} = \begin{cases} 1, & \text{if } \{v_i, v_j\} \in E, \\ 0, & \text{if } \{v_i, v_j\} \notin E. \end{cases}$$

The matrix depends on the ordering of the vertices, but it encodes the same graph. A matrix is *symmetric* when $a_{ij} = a_{ji}$ for all indices i, j . Its *diagonal entries* are $a_{11}, a_{22}, \dots, a_{nn}$.

- 1.5.** Using the order A, B, C, E, F, G, H , write the adjacency matrix of the Densmore graph. What do you observe about its diagonal and its symmetry? Explain why the sum of the entries in the row indexed by a vertex v is $d(v)$.

Part II — What would a single visit imply?

Suppose temporarily that every woman really visited the castle exactly once. Represent the stay of woman i by a closed time interval

$$I_i = [\ell_i, r_i] = \{t \in \mathbb{R} : \ell_i \leq t \leq r_i\},$$

where $\ell_i \leq r_i$, ℓ_i is her arrival time and r_i her departure time. Two intervals I_i and I_j *overlap* when their intersection is nonempty:

$$I_i \cap I_j \neq \emptyset.$$

Thus two women met if and only if their intervals overlap. The graph defined as follows is called their *intersection graph*: there is one vertex i for each interval I_i , and $\{i, j\}$ is an edge exactly when $I_i \cap I_j \neq \emptyset$. A graph that can be obtained in this way from intervals is called an *interval graph*.

Walks, paths, cycles and chords

A *walk of length k* is a finite sequence $(v_0, v_1, \dots, v_k)$ of vertices such that $\{v_{i-1}, v_i\} \in E$ for every $i = 1, \dots, k$. Its length is the number k of edges traversed. A *path* is a walk in which no vertex is repeated.

A *cycle of length $k \geq 3$* is a sequence of distinct vertices $(v_1, \dots, v_k)$ such that $\{v_i, v_{i+1}\} \in E$ for $i = 1, \dots, k-1$ and $\{v_k, v_1\} \in E$. Two vertices are *consecutive on the cycle* when they appear next to each other in this cyclic sequence. A *chord* is an edge joining two vertices of the cycle that are not consecutive on it.

A graph is *chordal* (or *triangulated*) if every cycle of length at least four has a chord. A cycle of length at least four with no chord is called a *chordless cycle*.

- 2.1.** We first establish a fact that will be used repeatedly. Let $W = (v_0, \dots, v_k)$ be a walk from v_0 to v_k .
- Suppose that $v_i = v_j$ for some $i < j$. Explain how to delete the part of the walk between these two occurrences while preserving the same first and last vertices.
 - Verify that the shortened sequence is still a walk.
 - Explain why repeating this operation must stop after finitely many steps.
 - Conclude that if a walk joins two vertices, then a path joins the same two vertices.

- 2.2.** Draw four intervals whose intersection graph is a path of length three. Then try to draw four intervals whose intersection graph is a chordless cycle of length four. What difficulty do you encounter?
- 2.3.** Consider any cycle $(v_1, \dots, v_k)$, with $k \geq 4$, in a graph obtained from intervals. Choose, among the intervals corresponding to the vertices of the cycle, an interval I_{v_i} whose right endpoint r_{v_i} is smallest.
- (a) Why does each of the two neighbours of v_i on the cycle have a left endpoint no greater than r_{v_i} ?
 - (b) Why does each of these two intervals have a right endpoint no smaller than r_{v_i} ?
 - (c) Deduce that the two neighbours of v_i meet each other.
 - (d) Why does this produce a chord?
- 2.4.** Conclude that if all seven women had made exactly one visit, the graph of encounters would necessarily be chordal. Why is the alternative name “triangulated” appropriate?

Part III — Solving the murder

Deleting a vertex

Let $x \in V$. The graph obtained by *deleting* x is denoted by $G - x$. Its vertex set is

$$V \setminus \{x\} = \{v \in V : v \neq x\},$$

and its edges are exactly the edges of G that do not have x as an endpoint. Thus deleting x also deletes every edge incident with x .

- 3.1.** Find a chordless cycle of length four in G . Find two more if possible.
- 3.2.** Explain why the woman who lied must belong to every chordless cycle: after removing her vertex, the remaining graph must be compatible with one interval per woman.
- 3.3.** Show that exactly one vertex belongs to all the chordless cycles you found.
- 3.4.** Verify that after deleting Ann, the graph $G - A$ is chordal. You may use the following strategy: identify vertices having only two neighbours in the remaining graph and check whether those two neighbours are adjacent.
- 3.5.** Is chordality alone sufficient to conclude that $G - A$ is an interval graph? Compare carefully the conclusion needed here with the implication proved in Part II.
- 3.6.** Construct explicitly a family of intervals whose intersection graph is $G - A$. Verify both the intersections corresponding to edges and the absence of intersections corresponding to non-edges.
- 3.7.** Who killed the Duke of Densmore?

What Problem 1 has added to our toolkit

We can now model a situation in which we record, for each pair of objects, whether or not a symmetric relationship holds. We distinguish the graph itself from any particular drawing, and we can encode it by an edge set or by an adjacency matrix. We can use neighbourhoods, degrees, paths, cycles and chords. We have also proved our first structural result:

Proposition 1. *The intersection graph of a finite family of intervals is chordal.*

2 How many relationships can a network contain?

Let $G = (V, E)$ be a finite simple undirected graph with $n = |V| \geq 1$ vertices and $m = |E|$ edges.

Part I — The densest possible graph

We shall establish the same result in three different ways. The first proof counts pairs, the second exploits the adjacency matrix, and the third uses mathematical induction. Each proof reveals a different idea that will be useful later.

We denote by $\binom{n}{2}$ the number of unordered pairs that can be formed from n distinct objects. The formula for this number will be derived in the first proof below.

4.1. First method: counting pairs. Suppose that the vertices represent n people and the edges represent mutual relationships.

- (a) How many ordered pairs (u, v) satisfy $u \neq v$?
- (b) Explain why every unordered pair $\{u, v\}$ produces exactly two of these ordered pairs: (u, v) and (v, u) .
- (c) Deduce, rather than assume, that

$$\binom{n}{2} = \frac{n(n-1)}{2}.$$

- (d) Explain why every edge is one of these unordered pairs, and why a simple graph has at most one edge for each pair. Conclude that

$$m \leq \binom{n}{2} = \frac{n(n-1)}{2}.$$

4.2. Second method: the adjacency matrix. Let $A = (a_{ij})$ be the adjacency matrix of G .

- (a) The entries *strictly above the diagonal* are those with indices $i < j$. How many such entries are there?
- (b) Show that

$$m = \sum_{1 \leq i < j \leq n} a_{ij} \quad \text{and} \quad 2m = \sum_{i=1}^n \sum_{j=1}^n a_{ij}.$$

Explain carefully why both identities are true.

- (c) Use either identity to prove again that $m \leq n(n-1)/2$.

4.3. Third method: induction. For $n \geq 1$, let $\mathcal{P}_n$ be the statement:

Every simple graph with n vertices has at most $n(n-1)/2$ edges.

- (a) Verify $\mathcal{P}_1$.
 - (b) Assume $\mathcal{P}_n$, and let G have $n + 1$ vertices. Remove one vertex v and all edges incident with it. How many edges can remain, by the induction hypothesis?
 - (c) How many edges incident with v can there be at most?
 - (d) Combine the two bounds and prove $\mathcal{P}_{n+1}$.
- 4.4. A graph is *complete* if every unordered pair of distinct vertices is an edge. The complete graph on n vertices is denoted by K_n . Prove that equality $m = n(n - 1)/2$ holds if and only if $G = K_n$. Explain how this equivalence appears in each of the three proofs.
- 4.5. Apply the result to the Densmore graph. How many edges could a graph on seven vertices have? How many does G actually have? What proportion of all possible edges are present?

Density

For a simple graph with $n \geq 2$ vertices, its *density* is the proportion of possible edges that are actually present:

$$\rho(G) = \frac{m}{\binom{n}{2}} = \frac{2m}{n(n-1)}.$$

The result proved above says that $0 \leq \rho(G) \leq 1$, and that $\rho(G) = 1$ if and only if $G = K_n$.

Part II — Counting the same objects twice

- 5.1. Imagine placing one token at each endpoint of every edge. How many tokens are placed in total when we count edge by edge?
- 5.2. How many tokens are placed at a fixed vertex v ? Count the total again, this time vertex by vertex.
- 5.3. Deduce the *handshake lemma*:
- $$\sum_{v \in V} d(v) = 2m.$$
- 5.4. Verify this identity for the Densmore graph using the degrees computed in Problem 1.
- 5.5. Split the sum of the degrees into the contribution of even-degree vertices and that of odd-degree vertices. Prove that every finite undirected graph has an even number of odd-degree vertices.
- 5.6. Can a graph have exactly 1, 3, 5 or 101 vertices of odd degree? Can it have exactly 100?

Part III — A first pigeonhole argument

The pigeonhole principle

If N objects are placed into K boxes and $N > K$, then at least one box contains at least two objects.

In the application below, the objects will be the n vertices and the boxes will be the possible values of their degrees. To use the principle rigorously, we must first determine how many degree values can actually occur in the same graph.

- 6.1. What are the possible degrees of a vertex in a simple graph with n vertices?

- 6.2.** Explain why a graph cannot simultaneously contain a vertex of degree 0 and a vertex of degree $n - 1$.
- 6.3.** Deduce that every simple graph with at least two vertices contains two vertices having the same degree.

Results established in Session 1

For every finite simple undirected graph $G = (V, E)$, with $n = |V|$ and $m = |E|$:

1. The intersection graph of a finite family of intervals is chordal.
2. $m \leq \binom{n}{2}$, with equality if and only if $G = K_n$.
3. If A is the adjacency matrix, then $m = \sum_{i < j} a_{ij} = \frac{1}{2} \sum_{i,j} a_{ij}$, and $d(v_i) = \sum_j a_{ij}$.
4. $\sum_{v \in V} d(v) = 2m$.
5. The number of odd-degree vertices is even.
6. If $n \geq 2$, two vertices have the same degree.

3 Homework — Connectivity and the number of edges

This section is not part of the three-hour session. Complete it after class, using the definitions and results established during the session. The questions form a self-contained investigation of connectedness and edge counts.

Connectivity

Two vertices u and v are *connected* if there exists a path whose first vertex is u and whose last vertex is v . A graph is *connected* if every pair of vertices is connected. It is *disconnected* if it is not connected.

For a vertex v , its *connected component* is the set

$$C(v) = \{u \in V : \text{there exists a path from } v \text{ to } u\}.$$

The structural properties of these sets will be proved in Question 7.1.

7.1. We first justify the use of the word *component*.

- (a) Why does $v \in C(v)$? Use a path of length 0.
- (b) If a path joins u to v , explain why reading it backwards gives a path from v to u .
- (c) Suppose that $u \in C(v)$ and $w \in C(u)$. Concatenate the two paths to obtain a walk from v to w , and use Question 2.1 to prove that $w \in C(v)$.
- (d) Deduce that if $u \in C(v)$, then $C(u) = C(v)$.
- (e) Deduce that two components are either equal or disjoint, and that every vertex belongs to exactly one component.
- (f) Give examples of connected and disconnected graphs with the same order, and list their components.

7.2. Let G be disconnected. Choose a component C having the smallest possible number of vertices, and write $s = |C|$.

- (a) Explain why G has at least two components and deduce that $1 \leq s \leq n/2$.
- (b) How many unordered pairs have one endpoint in C and the other in $V \setminus C$? Justify your count.
- (c) Why can none of these pairs be an edge?
- (d) Prove that $s(n-s) \geq n-1$ by verifying the identity

$$s(n-s) - (n-1) = (s-1)(n-s-1)$$

and explaining why its right-hand side is nonnegative.

- (e) Deduce that

$$m \leq \binom{n}{2} - (n-1) = \binom{n-1}{2}.$$

7.3. Prove the following sufficient condition:

$$m > \binom{n-1}{2} \implies G \text{ is connected.}$$

- 7.4. Show that the bound is sharp by constructing a disconnected graph with exactly $\binom{n-1}{2}$ edges.
- 7.5. Starting from one vertex of a connected graph, repeatedly add a new vertex joined by an edge to the vertices already reached. Explain why this can be continued until all vertices have been reached, and deduce the necessary condition

$$G \text{ connected} \implies m \geq n - 1.$$

- 7.6. For $n \geq 4$, construct both a connected graph and a disconnected graph having exactly $n - 1$ edges. Conclude that the number of edges alone cannot give a necessary and sufficient condition for connectivity.

Results established in the homework

For every finite simple undirected graph $G = (V, E)$, with $n = |V|$ and $m = |E|$:

1. The connected components form a partition of V .
2. If G is disconnected, then $m \leq \binom{n-1}{2}$.
3. If $m > \binom{n-1}{2}$, then G is connected; this threshold is sharp.
4. If G is connected, then $m \geq n - 1$.
5. For $n \geq 4$, the number of edges alone does not characterise connectivity.